



CityLearn Challenge:

District Load Control for the Texas Climate

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Abstract

This study investigates district-level load tracking in a residential building portfolio using the CityLearn Texas climate scenario. The objective is to coordinate battery storage and cooling demand to track a prescribed district load profile while satisfying operational constraints. A range of control architectures was systematically evaluated, encompassing rule-based heuristics, model predictive control (MPC), machine learning (ML)-based prediction, and reinforcement learning (RL) paradigms. Rule-based controllers served as baselines, while multiple MPC formulations were evaluated to investigate the impact of predictive modelling and coordination on load-tracking performance. In parallel, RL agents leveraging the Soft Actor-Critic algorithm were developed to explore both decentralized and hierarchical coordination topologies. To support the learning and control frameworks, a feature engineering pipeline was developed that transformed raw observations into building-level, portfolio-level, and temporal representations. These representations underpinned the development of an initial ML predictive controller, a behaviour-based building clustering methodology, and a hybridized group-aware controller. The latter synergized ML-derived action proposals with proportional-integral (PI) feedback correction and cluster-specific action redistribution. Results show that performance improved with increasing predictive capability and coordination. While rule-based controllers exhibited the weakest load-tracking performance, MPC-based approaches achieved substantial improvements, with the HRN-MPC controller emerging as the strongest MPC architecture. The hierarchical reinforcement learning controller (Hier-2) achieved the best overall load-tracking performance, reducing the CV-RMSE to 34.5% and outperforming all evaluated MPC, ML-based, and rule-based approaches. The ML-based experiments further showed that prediction alone is insufficient for reliable closed-loop control, whereas the integration of feedback correction and coordinated action redistribution significantly improves performance. Ultimately, this study highlights the importance of prediction, adaptation, and hierarchical coordination in advancing smart district energy management and demand response applications.

Index Terms— Demand Response, Energy Flexibility, Smart Buildings, District Energy Management, Load Tracking, Model Predictive Control, Reinforcement Learning, Hierarchical Control, CityLearn, Proportional-Integral (PI) Control, Building Clustering, Coordinated Control, Closed-Loop Control, Multi-Agent Systems, Battery Energy Storage Systems (BESS), Grid-Interactive Efficient Buildings (GEB), Peak Shaving, Electricity Markets, Transactive Energy, Distribution Shift.

ABBREVIATIONS

Abbreviation	Meaning
AL-MPC	Adaptive LSTM-Linearized Model Predictive Control
ANN	Artificial Neural Network
BESS	Battery Energy Storage System
CJSS-MPC	Coordinated Joint State-Space Model Predictive Control
CV-RMSE	Coefficient of Variation of the Root Mean Squared Error
DDPG	Deep Deterministic Policy Gradient
DG	Distributed Generation
DR	Demand Response
DSO	Distribution System Operator
DQN	Deep Q-Network
DRL	Deep Reinforcement Learning
EV	Electric Vehicle
HVAC	Heating, Ventilation and Air Conditioning
HRN-MPC	Hybrid Residual-NARX Model Predictive Control
LSTM	Long Short-Term Memory
MARL	Multi-Agent Reinforcement Learning
MDP	Markov Decision Process
MPC	Model Predictive Control
NARX	Nonlinear AutoRegressive with eXogenous Inputs
NMBE	Normalised Mean Bias Error
PPO	Proximal Policy Optimization
PV	Photovoltaic
RBC	Rule-Based Control
RES	Renewable Energy Sources
RL	Reinforcement Learning
SAC	Soft Actor-Critic
SoC	State of Charge
TD	Temporal Difference
TOU	Time-of-Use

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I. INTRODUCTION

The Netherlands ranks among the European countries with the highest per-capita installed capacity of rooftop solar photovoltaic (PV) systems [1]. For more than two decades, Dutch households benefited from the *salderingsregeling*, a net metering policy that allowed surplus PV generation to be offset against grid consumption. As the policy is phased out and scheduled for full abolition in 2031, the timing of electricity storage and consumption is becoming increasingly important for households with rooftop PV [2].

Simultaneously, the rapid growth of distributed solar generation has contributed to severe grid congestion (*netcongestie*) across large parts of the Dutch distribution network [3], [4]. Distribution system operators (DSOs) are refusing new PV connection requests in affected areas because the low-voltage infrastructure cannot absorb the simultaneous midday generation peaks produced by co-located rooftop installations [3]. Figure 1 illustrates the current extent of congestion across the Dutch grid. The primary tool that DSOs are deploying to address congestion is *demand response*: incentivising buildings and aggregators to reshape their electricity consumption in line with real-time grid conditions, thereby reducing the need for costly network reinforcement [5].

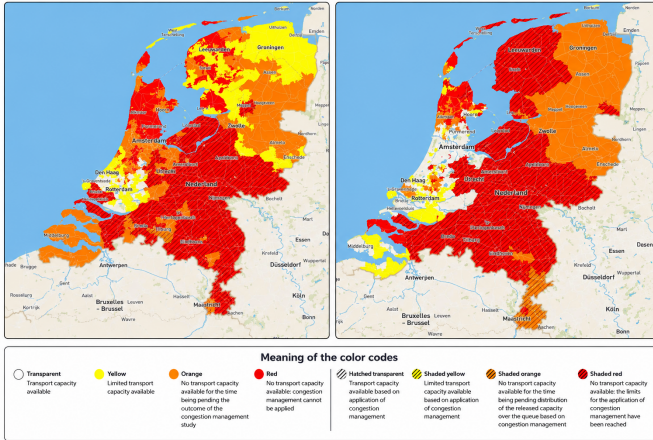


Fig. 1. Grid congestion map of the Dutch electricity distribution network. Shaded areas indicate regions where new connection requests are being refused or delayed [3].

These developments reflect a broader global trend: buildings are no longer passive loads but must become active, grid-interactive assets capable of delivering flexibility services. Globally, the buildings and construction sector accounted for approximately 34 per cent of energy-related CO₂ emissions and over 32 per cent of total energy demand in 2023 [6]. Coordinating building-level distributed energy resources, battery energy storage systems (BESS), heat pumps, and PV generation, to follow a prescribed grid-level target is therefore both an energy and a climate imperative [7], [8].

Several control paradigms have been proposed for district-level energy management, including rule-based control (RBC), model predictive control (MPC), machine learn-

ing (ML)-based approaches, and reinforcement learning (RL). RBC is straightforward to deploy but relies on predefined heuristics that cannot adapt to changing operating conditions. MPC incorporates predictions of future system behaviour and can achieve near-optimal control performance, but its effectiveness depends heavily on the availability of accurate system models. ML and RL approaches offer data-driven alternatives that can learn complex control strategies directly from historical data or interaction with the environment, reducing the need for explicit physical modelling [5], [8].

While each of these paradigms has demonstrated promise for building energy management, their relative strengths and limitations for district-level load tracking remain an active area of research. In particular, it remains unclear how predictive capability, adaptation, and coordination influence the trade-off between load-tracking accuracy and occupant comfort across different controller architectures.

This paper investigates district-level load tracking for a portfolio of 25 residential buildings equipped with BESS and rooftop PV within the IEA EBC Annex 96 Common Exercise 1 framework. A broad range of control approaches are evaluated, including RBCs, MPCs, ML-based methods, and RL architectures, spanning a range of complexity from simple heuristic controllers to predictive and hierarchical control strategies. Through a systematic comparison, the study examines how different control architectures influence load-tracking performance, thermal comfort, and district-level coordination.

From a societal perspective, the findings highlight the potential of intelligent energy management systems to provide flexibility services to the grid. As the phase-out of the *salderingsregeling* and increasing grid congestion place additional pressure on electricity networks, effective coordination of building loads, battery storage systems, and PV generation can play an important role in supporting future demand response programs.

Research Questions

The central question motivating this study is: *Which control architecture best enables a district of residential buildings to track a prescribed daily load profile while maintaining occupant comfort, and which design choices drive that performance?*

- 1) Which coordination mechanisms between building-level controllers are effective at meeting a daily portfolio-level target profile, compared to non-coordinated control?
- 2) How do key constraints and objective weights (e.g., comfort bounds and battery limits) influence performance and coordination behaviour?
- 3) Among the evaluated approaches, which control system achieves the best trade-off between reference tracking accuracy (CV-RMSE) and temperature comfort, and under what conditions does it not meet the tracking requirements?

II. LITERATURE REVIEW

A. Energy Flexibility and Demand Response in Buildings

The transition towards a low-carbon electricity system requires buildings to evolve from passive energy consumers into active, grid-interactive assets [7], [8]. Energy flexibility, the ability to modify the magnitude, timing, or duration of electricity demand in response to grid signals, is central to this transition [7]. Li et al. distinguish between several flexibility modes: load shifting, load shedding, modulation, and on-site generation dispatch. These modes must be aggregated across building clusters or district scale to have meaningful impact on grid operations, since single-building flexibility is too small for transmission-level grid services [7]. Demand response (DR), the event-driven reduction or shift of consumption in response to price or reliability signals, is the most widely deployed form of this flexibility [9]. Critically, Li et al. identify hierarchical and coordinated control architectures as the most practical balance between performance and scalability, ahead of fully centralised or purely price-responsive decentralised approaches [7].

B. Reinforcement Learning for Building Energy Control

Model predictive control (MPC) has long been the preferred approach for building energy optimisation, but it requires accurate physics-based models of each building that are expensive to develop and maintain at district scale [8]. Reinforcement learning (RL) overcomes this barrier by learning effective control policies through direct interaction with the environment, without an explicit building model [5], [8]. Nagy et al. identify RL as particularly well-suited to problems involving sequential decision-making over long time horizons, such as battery pre-positioning ahead of demand peaks, where rule-based control falls short [8]. Vázquez-Canteli and Nagy conducted a comprehensive review of RL algorithms for demand response and concluded that single-agent RL is adequate for price-response in individual buildings, but that multi-agent coordination is essential once multiple buildings interact, since independent agents can merely relocate peak demand rather than reduce it [5]. Furthermore, incorporating future information, such as load targets and weather forecasts, into the agent's observation has been shown to substantially improve control performance by enabling proactive rather than reactive decisions [10].

C. Multi-Agent Coordination and Control Architectures

The choice of coordination architecture plays a critical role in district-level energy management and reinforcement learning performance [5], [8]. Three architectures are commonly considered: centralized, decentralized, and distributed control. In a centralized architecture, a single controller determines the actions of all buildings using district-wide information, enabling direct coordination but suffering from scalability challenges as the number of buildings increases [11]. In contrast, decentralized architectures assign an independent controller to each building, improving scalability but limiting coordination between buildings [5].

Distributed architectures provide a compromise between these two extremes. Buildings maintain local controllers while exchanging selected information through communication or coordination mechanisms. This enables cooperation while avoiding the computational burden associated with fully centralized control. A common approach is Centralized Training with Decentralized Execution (CTDE), where agents learn cooperative behaviours during training but operate independently during deployment [8].

Recent studies further demonstrate the importance of coordination architecture design. For example, the MERLIN framework combines decentralized SAC agents with transfer learning to improve district-level performance while reducing training requirements [12].

D. The CityLearn Simulation Platform

CityLearn is an open-source simulation platform designed for evaluating and comparing energy management controllers in building districts [13]. The platform provides a standardized benchmark environment that supports rule-based control (RBC), model predictive control (MPC), and reinforcement learning (RL), enabling different control strategies to be evaluated under the same conditions. The version used in this work includes data-driven building thermal models, battery energy storage systems, electric vehicle integration, weather forecasts, and realistic occupant behaviour, allowing controllers to be tested in scenarios that closely resemble real residential energy systems [14].

Previous studies have demonstrated the suitability of CityLearn for benchmarking advanced control strategies. Results from the CityLearn Challenge and subsequent studies showed that both RL-based and model-based approaches can achieve competitive performance and that RL often outperforms rule-based control for several energy management objectives [13], [15]. These findings establish CityLearn as an appropriate benchmark for evaluating district-level load tracking and demand response strategies.

Collectively, the reviewed literature highlights the importance of prediction, coordination, and scalable control architectures for achieving effective district-level energy management.

E. Research Gap

The reviewed literature highlights the importance of prediction, coordination, and scalability in district-level energy management. However, important challenges remain regarding the design of effective coordination strategies for multi-building systems, while systematic studies investigating the impact of design choices such as forecast horizon, observation space, network capacity, and action scaling on load-tracking performance are limited. The present study addresses these gaps through a comparative evaluation of RBC, MPC, ML-based, and RL-based controllers within the IEA EBC Annex 96 CityLearn benchmark.

III. METHODOLOGY

This section presents the methodology used in this study. First, the dataset and feature engineering process are described, followed by the implementation of the evaluated control approaches.

A. Dataset

The experiments in this work were conducted using the Texas climate dataset, which represents a district of 25 residential buildings located in a cooling-dominated region. Each building is equipped with rooftop photovoltaic (PV) generation and a battery energy storage system (BESS), providing a realistic benchmark for district-level energy management and demand response studies.

The dataset contains one year of hourly data describing building thermal conditions, electricity demand, HVAC operation, renewable generation, and weather conditions. In addition to current weather measurements, short-term forecasts of outdoor temperature, relative humidity, and solar irradiance are available, enabling the development of predictive control strategies.

Following the benchmark specification, June is used for controller development and July for evaluation. District-level target load profiles are only available for these two months, making them the exclusive periods considered in this study.

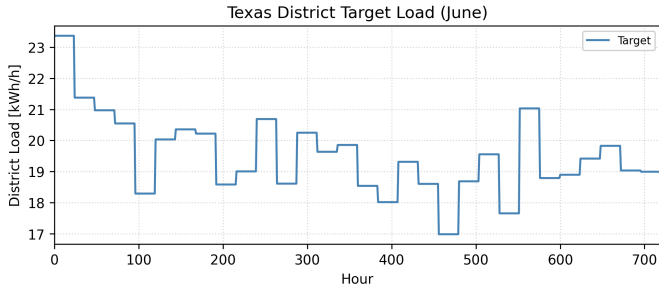


Fig. 2. Daily district-level target load profile used in the Texas climate scenario for the entire month of June.

A distinctive characteristic of the Texas dataset is the target load profile shown in Figure 2. Rather than varying continuously throughout the day, a single target value is assigned to each 24-hour period. Consequently, the desired district load remains constant within a given day and changes only between consecutive days, encouraging stable district-level electricity consumption around a daily reference value.

Together, the building dynamics, renewable generation, energy storage systems, weather variability, and district-level targets provide a comprehensive benchmark for evaluating advanced control strategies in multi-building energy management systems.

B. Feature Engineering

Before developing the final group-aware controller, feature engineering was performed to convert the raw

CityLearn observations into compact machine-learning features. CityLearn provides a benchmark environment for evaluating grid-interactive building control strategies [14]. In the Texas environment, each building observation contains approximately 29 variables. Since the district contains 25 buildings, a flattened centralised observation would contain approximately $25 \times 29 = 725$ scalar values per timestep. Directly using this high-dimensional raw representation would make the later models less interpretable and harder to analyse. Therefore, the raw observations were transformed into building-level, portfolio-level, and time-dependent features.

Time-dependent features were included because electricity demand and cooling behaviour follow strong daily patterns. Instead of using only the raw hour value, cyclic sine and cosine encodings were used:

$$h_{\sin} = \sin\left(\frac{2\pi h}{24}\right), \quad h_{\cos} = \cos\left(\frac{2\pi h}{24}\right),$$

where h is the hour of day. This avoids the artificial discontinuity between hour 23 and hour 0 and helps the model represent repeated daily load shapes, night-time demand reduction, and afternoon cooling peaks.

Building-level features were generated to describe the behaviour of each individual building. These included information related to net electricity consumption, battery state of charge, indoor temperature, cooling and heating setpoints, cooling demand, and solar generation. Such features are important because the 25 buildings do not respond identically to the same control action. For example, a building with a low battery state of charge may not be suitable for discharging, while a building that is already below its cooling setpoint should not receive additional cooling action.

Portfolio-level features were generated to represent the aggregate district behaviour. Since the objective is district-level target tracking, the most important portfolio feature is the tracking error:

$$e_t = d_t^* - L_t,$$

where d_t^* is the district target and L_t is the current portfolio load. A positive error indicates that the portfolio load is below the target, while a negative error indicates that the portfolio load is above the target. This error later became an important input to the PI feedback layer of the final controller.

In addition, an action-response dataset was generated by applying random battery and cooling actions in the CityLearn environment and recording the resulting portfolio response. This dataset was used to train a response model based on gradient boosting, implemented using scikit-learn, to predict the next portfolio load from the engineered state and candidate actions [16], [17]. The trained response model formed the predictive component of the initial ML controller and the final group-aware controller.

C. Rule-Based Controllers (RBCs)

Rule-based controllers (RBCs) available within the CityLearn framework were evaluated as baseline control strategies for comparison against the proposed more advanced

controller approaches. These controllers rely on predefined heuristics rather than learning from interaction with the environment. Their inclusion provides a reference point for assessing the performance gains achieved by more advanced controllers.

The evaluated RBCs can be broadly categorized into two groups. The first group consists of time-scheduled controllers that determine battery charging, discharging, and HVAC operation according to the hour of the day. The second group consists of temperature-responsive controllers that use indoor temperature information and thermal comfort setpoints to regulate HVAC operation. All controllers were evaluated under both centralized and decentralized control configurations. Table I summarizes the main characteristics of the evaluated controllers.

TABLE I
RULE-BASED CONTROLLERS EVALUATED IN THIS WORK

Controller	Control Strategy
HourRBC	Generic hour-based controller. Actions are determined solely from the current hour according to a predefined 24-hour action schedule.
BasicRBC	Charges the battery by 9.1%/h during 22:00–08:00 and discharges by 8.0%/h during 09:00–21:00. Cooling power is reduced at night (40%) and increased during daytime hours (80%).
OptimizedRBC	Grid-search optimized schedule. Battery actions vary throughout the day: +5.53%/h (01:00–06:00), +3.4%/h (23:00–24:00), and discharge rates between -2.0%/h and -4.4%/h during daytime periods.
BasicBatteryRBC	Solar-oriented strategy that charges the battery by 11.0%/h between 06:00–14:00 to absorb PV generation and discharges by 6.7%/h during all remaining hours.
TemperatureBasedRBC	HVAC control is determined by the instantaneous temperature error relative to the comfort setpoint. Larger temperature deviations result in stronger cooling actions, while battery operation follows a predefined schedule.
PITemperatureController	Extends the TemperatureBasedRBC by incorporating proportional-integral (PI) feedback. HVAC actions depend on both the current temperature error and the accumulated error over time, while battery scheduling remains unchanged.

The BasicRBC, OptimizedRBC, and BasicBatteryRBC primarily focus on energy shifting through predefined battery schedules. In contrast, the TemperatureBasedRBC and PITemperatureController additionally use thermal state information to regulate HVAC operation. Together, these controllers provide a diverse set of baseline strategies ranging from simple hourly schedules to feedback-based temperature control. Evaluating their performance under both centralized and decentralized control architectures establishes a comprehensive benchmark against which the proposed advanced controllers can be assessed.

D. Model Predictive Controllers (MPCs)

Model Predictive Control (MPC) determines control actions by repeatedly solving an optimization problem over a finite prediction horizon using a predictive model of the

building dynamics. At each timestep, only the first action of the optimal sequence is applied before the optimization is repeated using updated measurements. Since MPC performance depends strongly on prediction accuracy, each controller incorporates a System Identification (SID) stage to model building behaviour.

TABLE II
SUMMARY OF EVALUATED MPC ARCHITECTURES

Controller	SID Model	Key MPC Feature
AL-MPC	CityLearn LSTM model with local linearization	Decentralized MPC using distributed district targets
HRN-MPC	Hybrid Residual-NARX model	Data-driven thermal prediction with residual learning
CJSS-MPC	Joint state-space residual-NARX model	District-level coordination through flexibility allocation

1) *Adaptive LSTM-Linearized MPC (AL-MPC)*: AL-MPC directly utilizes the pre-trained CityLearn dynamics model. To enable computationally efficient optimization, the nonlinear model is locally linearized around the current operating point at every timestep:

$$x_{k+1} = A_k x_k + B_k u_k + E_k d_k.$$

A decentralized MPC controller is assigned to each building, while the district target is distributed according to estimated baseline loads. The baseline load represents the expected electricity consumption of a building in the absence of additional control actions. Buildings with larger expected demand are therefore assigned a larger share of the district target, allowing the aggregate district load to collectively follow the desired profile.

The controller simultaneously manages battery charging/discharging and cooling operation. The optimization objective balances load tracking, thermal comfort, action smoothness, and battery state-of-charge regulation:

$$\min_u J_{\text{track}} + J_{\text{comfort}} + J_{\text{smooth}} + J_{\text{SOC}}.$$

2) *Hybrid Residual-NARX MPC (HRN-MPC)*: HRN-MPC employs a fully data-driven SID framework. Building dynamics are represented using a NARX model trained from randomized excitation data. A linear regression model captures the dominant system behaviour, while a residual neural network learns the remaining nonlinear prediction error. The resulting thermal model is expressed as

$$T_{k+1} = f(T_k, \dots, T_{k-n}, u_k, \dots, u_{k-m}, d_k).$$

The identified model predicts future indoor temperature using both current operating conditions and historical system behaviour. This enables the controller to capture delayed thermal responses, where the effects of cooling actions may only become visible several timesteps later.

The learned model is incorporated into a decentralized MPC formulation that minimizes

$$\min_u J_{\text{track}} + J_{\text{comfort}} + J_{\text{effort}} + J_{\text{SOC}}.$$

3) *Coordinated Joint State-Space MPC (CJSS-MPC)*: CJSS-MPC extends HRN-MPC by jointly modelling multiple building states rather than only indoor temperature. The learned state vector includes variables such as indoor temperature, battery state-of-charge, cooling electricity consumption, and net building electricity demand:

$$x_{k+1} = f(x_k, u_k, d_k).$$

By modelling both thermal and electrical states simultaneously, the controller can explicitly capture the interactions between HVAC operation, battery behaviour, and electricity consumption. This provides a more comprehensive representation of building dynamics than a temperature-only model.

A supervisory coordination layer operates above the local building controllers. At each timestep, the coordinator estimates future district demand and distributes the required correction among buildings according to their estimated flexibility. Flexibility refers to the ability of a building to modify its electricity consumption while remaining within operational and comfort constraints. The target allocation mechanism is represented as

$$P_{\text{target}}^{(i)} = P_{\text{baseline}}^{(i)} + \alpha_i (P_{\text{district,target}} - P_{\text{district,baseline}}).$$

where α_i denotes the flexibility score assigned to building i . Buildings with greater available battery capacity or thermal margin are assigned a larger share of the required load adjustment. Each building subsequently solves a local MPC problem while tracking its allocated target:

$$\min_u J_{\text{track}} + J_{\text{comfort}} + J_{\text{ramp}} + J_{\text{smooth}} + J_{\text{SOC}}.$$

This hierarchical structure enables cooperation between buildings while preserving the scalability advantages of decentralized control.

E. Initial ML Predictive Controller Performance

The initial ML predictive controller was evaluated in the live CityLearn environment to test whether the offline-trained response model could be used directly for closed-loop district load tracking [14]. The response model was based on gradient boosting and implemented using scikit-learn [16], [17]. The controller achieved an NMBE of 67.8426% and a CV-RMSE of 146.3904%. The large positive NMBE indicates a strong systematic tracking bias, while the high CV-RMSE shows that the controller was unable to track the target reliably during live operation.

Figure 3 shows the controller behaviour over the first 3 days. Although the ML-predicted next load remains relatively close to the district target, the actual portfolio load exhibits

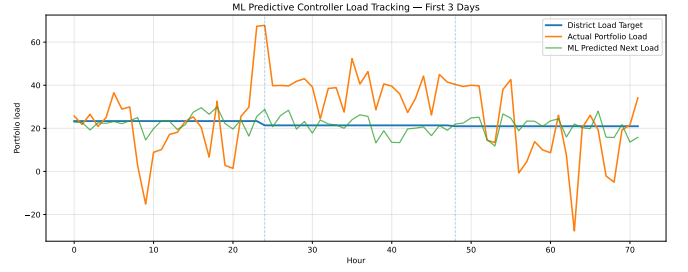


Fig. 3. District target, actual portfolio load, and ML-predicted next load for the initial ML predictive controller over the first 3 days of operation. Although the predicted load remains relatively close to the target, the actual portfolio load exhibits large oscillations, indicating model mismatch and closed-loop distribution shift.

large oscillations and frequent deviations. This mismatch between predicted and realised behaviour indicates closed-loop distribution shift: the model was trained on random-action data, but in live control the future states are influenced by the controller's own actions.

These results show that the ML response model was not sufficient on its own for reliable target tracking. This motivated the final group-aware controller, which combines the ML proposal with PI feedback correction and cluster-aware redistribution.

F. Building Clustering Results

Figure 4 shows the silhouette score for different numbers of building clusters [18]. Although two clusters achieved the highest score, five clusters were selected for the final controller to provide finer control granularity across the 25-building portfolio. This allowed the controller to redistribute actions across more specific building groups while still keeping the number of groups manageable, consistent with hierarchical community-level coordination approaches in building energy control [19].

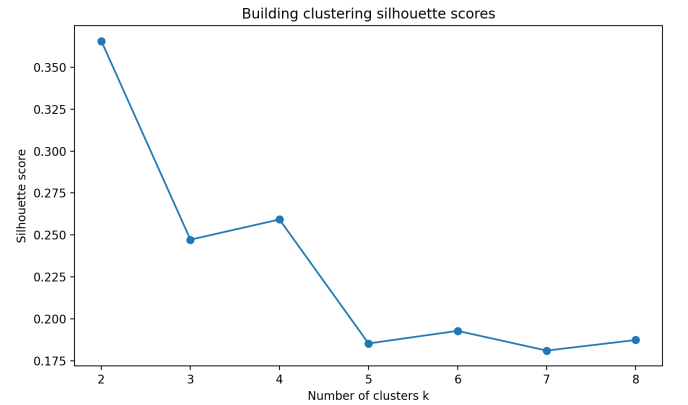


Fig. 4. Silhouette scores for different numbers of building clusters. Although two clusters achieved the highest silhouette score, five clusters were selected for the final controller to provide finer control granularity across the 25-building portfolio.

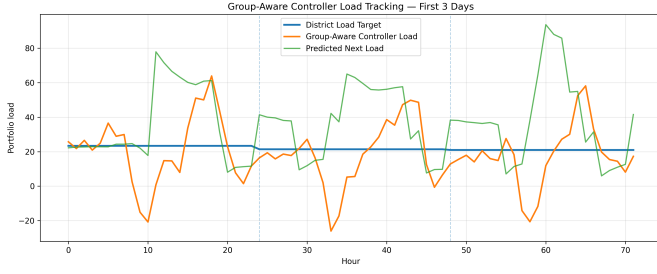


Fig. 5. District target, group-aware controller load, and predicted next load over the first 3 days of operation. This diagnostic view shows the short-term closed-loop behaviour after adding PI feedback correction and cluster-aware action redistribution.

G. Final Group-Aware Controller Performance

Figure 5 shows the short-term closed-loop behaviour of the final group-aware controller over the first 3 days. The actual portfolio load still exhibits noticeable hour-to-hour oscillations, indicating that the multi-building system remains difficult to control precisely. However, compared with the initial ML predictive controller, the final controller uses PI feedback and cluster-aware redistribution to reduce the systematic tracking bias caused by relying on the learned response model alone. This follows the broader idea that hierarchical and coordinated control architectures can help balance district-level objectives with building-level heterogeneity in grid-interactive building control [19].

The predicted next load does not always match the realised portfolio load, showing that model mismatch is still present during closed-loop operation. Nevertheless, the feedback and redistribution layers help reduce the systematic bias observed in the pure ML controller. This supports the main motivation for the final architecture: prediction alone is not sufficient, but combining prediction with feedback correction and group-aware redistribution produces a more robust controller.

H. Reinforcement Learning Controllers (RLs)

1) *Soft Actor-Critic*: Soft Actor-Critic (SAC) [20] is an off-policy actor-critic algorithm for continuous action spaces that maximizes both expected cumulative reward and the entropy of the policy:

$$J(\pi) = \sum_t \mathbb{E}[r(s_t, a_t) + \alpha \mathcal{H}(\pi(\cdot | s_t))],$$

where $\mathcal{H}(\pi(\cdot | s_t))$ is the Shannon entropy of the action distribution at state s_t and α is a temperature parameter that controls the exploration–exploitation trade-off. SAC maintains two Q-networks Q_{ϕ_1}, Q_{ϕ_2} with corresponding target networks to reduce overestimation bias, and a stochastic policy network $\pi_\theta(a | s)$ that outputs a squashed Gaussian distribution over actions. The policy is updated to minimize:

$$J_\pi(\theta) = \mathbb{E}_{s, a \sim \pi_\theta} [\alpha \log \pi_\theta(a | s) - \min_i Q_{\phi_i}(s, a)].$$

Entropy regularization improves training stability and prevents early convergence, making SAC well-suited to building energy

control tasks where the reward landscape is non-stationary across episodes [8].

2) *Centralized vs. Decentralized Formulation*: Both centralized and decentralized formulations of SAC were implemented and evaluated as initial controllers. In the centralized formulation, a single agent observes the linked state of all 25 buildings and outputs a joint battery action vector. In the decentralized formulation, one independent SAC agent is trained per building on its own local observation. The decentralized variant produced substantially better load-tracking performance in early experiments, consistent with the finding of Nweye et al. that per-building policies can adapt to individual occupant patterns and outperform a shared central policy at the district level [12]. All subsequent development therefore focused on the decentralized paradigm.

3) *Decentralised SAC for Battery Control*: Each of the 25 building agents is trained independently within the CityLearn environment [14]. Rather than operating on the raw 29-dimensional CityLearn observation, every agent receives the 9-dimensional feature-engineered state vector as described in Section III-B. Cooling-device actions are deactivated for all buildings, restricting each agent to a scalar battery action where negative values correspond to charging and positive values to discharging. Agents are trained with a shared tracking reward that penalizes the squared deviation of the district aggregate load from the prescribed target profile. After training, the policy and Q-network weights of all 25 agents, together with their observation-normalization statistics, are saved and subsequently used as frozen lower-level components in the hierarchical controller described below.

4) *Hierarchical SAC Architecture (Hier-2)*: The Hier-2 controller extends the decentralized agents by placing a single district-level coordinator on top of the frozen lower-level policies. The frozen lower-level agents propose nominal battery actions based on their building-level observations, and the upper-level SAC agent applies a small corrective offset to steer the aggregate district load toward the daily target profile. The 25 building-level SAC agents trained in the preceding stage are held fixed throughout upper-level training to prevent interference between the two learning objectives and reduces the effective state and action spaces that the upper-level agent must explore.

A single upper-level SAC agent coordinates the lower-level specialists at the district level. Its observation vector $s_t \in \mathbb{R}^9$ comprises $\sin(2\pi h_t/24)$ and $\cos(2\pi h_t/24)$ (time-of-day encoding, where h_t is the hour of day), $L_{t-1}/30$ (previous district load, normalized), $d_t^*/30$ (current load target, normalized), $(L_{t-1} - d_t^*)/30$, $\bar{s}_t$ (mean state of charge across all 25 buildings), $d_{t+6}^*/30$ (6-hour-ahead load target), $d_{t+12}^*/30$ (12-hour-ahead load target) and $(d_{t+6}^* - d_t^*)/10$ (normalized target slope). The 6-hour and 12-hour look-ahead targets enable the coordinator to anticipate future demand shifts and pre-position battery storage accordingly, distinguishing Hier-2 from reactive single-step controllers. The policy network consists of two fully connected hidden layers with 64 units each and ReLU activations. SAC hyperparameters are: learning rate 3×10^{-4} ,

discount factor $\gamma = 0.99$, soft target update rate $\tau = 0.005$, and entropy temperature $\alpha = 0.2$.

The upper-level agent outputs a scalar $u_t \in [-1, 1]$, which is converted into a per-building corrective battery offset through three sequential transformations. First, the action is scaled by 0.35 to limit the maximum per-step perturbation. Second, an exponential moving average (EMA) with smoothing factor 0.4 suppresses rapid reversals that would otherwise produce oscillatory district-load behavior:

$$\Delta_t = 0.4 r_{\text{raw}} + 0.6 \Delta_{t-1}.$$

Third, the smoothed correction is added to each lower-level action and clipped to the feasible range:

$$a'_i = \text{clip}(a_i + \Delta_t, -1, 1).$$

The upper-level agent receives the composite reward:

$$r_t = -0.01 (L_t - d_t^*)^2 - 0.05 \Delta_t^2,$$

where the first term penalizes deviation from the target and the action-penalty term discourages large corrections that could disrupt the lower-level agents' learned policies.

5) *Hier-2 Variants*: Because Hier-2 achieved the best tracking accuracy among all evaluated controllers, five variants were developed to investigate whether targeted modifications to the observation space, forecast horizon, SoC representation, network capacity, or action scaling could yield further improvement. Table III summarizes each variant with its design rationale and anticipated trade-offs.

TABLE III
HIER-2 VARIANTS: DESIGN MODIFICATIONS AND TRADE-OFFS

Variant	Advantage	Disadvantage
Hier-2.1-Solar 12-dim obs; adds district solar generation	Enables proactive charging ahead of solar-driven load drops	Solar contribution is already implicit in the net load signal
Hier-2.2-Extended 12-dim obs; extends forecast to 18 h and 24 h; adds mean indoor temperature	Longer planning horizon captures the full daily demand cycle	Distant targets add noise to the learning signal
Hier-2.3-SOC 11-dim obs; replaces mean SoC with mean, std, and min SoC	Upper agent can detect heterogeneous battery states	SoC spread adds little information when all agents share the same policy structure
Hier-2.4-LargeNet same 9-dim obs; policy networks [128, 128, 64]	Higher representational capacity for complex coordination patterns	Larger networks require more training episodes to converge
Hier-2.5-Asymmetric same 9-dim obs; discharge scale 0.50, charge scale 0.25	Biases coordinator toward discharging during over-supply periods	Asymmetric scaling conflicts with symmetric demand patterns

6) *Training Procedure*: The 25 building-level SAC agents were trained for 100 episodes on a 720-step simulation of the TX neighbourhood (dataset hours 3 624–4 343). Each agent updated its Q-networks and policy after every environment step using mini-batches of 128 transitions sampled uniformly from a per-agent replay buffer with a capacity of 72 000 transitions. Observation normalization was activated once 512 transitions had been collected. Total wall-clock training time was approximately 9.4 hours on CPU. The upper-level coordinator was trained for 100 episodes over the same simulation window, with the lower-level agents frozen. At each step, the frozen lower-level agents proposed nominal battery actions; the upper-level agent computed and applied an EMA-smoothed correction and the CityLearn environment advanced by one hour. Gradient updates used mini-batches of 128 transitions from a replay buffer of 72 000 transitions.

I. Evaluation Metrics (KPIs)

Controller performance was evaluated using three key performance indicators (KPIs) that assess both load-tracking accuracy and building thermal comfort.

Normalized Mean Bias Error (NMBE) measures whether the district load is, on average, above or below the target load.

$$\text{NMBE} = \frac{\frac{1}{N} \sum_{t=1}^N (y_t - y_t^{\text{ref}})}{\frac{1}{N} \sum_{t=1}^N y_t^{\text{ref}}} \times 100$$

where y_t is the district net load and y_t^{ref} is the target load. Values closer to zero indicate better tracking performance.

Coefficient of Variation of the Root Mean Squared Error (CV-RMSE) measures the overall tracking error between the district load and the target load. Lower values indicate more accurate target tracking.

$$\text{CV-RMSE} = \frac{\sqrt{\frac{1}{N} \sum_{t=1}^N (y_t - y_t^{\text{ref}})^2}}{\frac{1}{N} \sum_{t=1}^N y_t^{\text{ref}}} \times 100$$

Temperature Comfort Violation measures the percentage of time indoor temperatures fall outside the acceptable comfort range for the Texas climate, which is 22°C–26°C:

$$\text{Temp Comfort Violation} = \frac{1}{N} \sum_{t=1}^N \mathbf{1}(T_t \notin [22, 26]) \times 100$$

where T_t denotes the indoor temperature at timestep t . Lower values indicate better maintenance of thermal comfort.

IV. RESULTS

This section reports the evaluation results for each methodological component in the order introduced in Section III.

A. Feature Engineering

The feature engineering stage produced building-level and portfolio-level datasets that were later used for clustering, response-model training, and controller development. The building-level features captured individual building behaviour, while the portfolio-level features represented aggregate district target-tracking conditions.

Principal component analysis was used as a diagnostic measure of feature compactness. The first three building-level principal components explained 58.38% of the variance, while the first three portfolio-level components explained 68.96%. This indicates that the portfolio-level features were more compact, whereas the building-level features contained more diverse behaviour across the individual buildings.

An action-response dataset was then generated using random battery and cooling actions. It contained 17,280 samples over 8 episodes and used `next_portfolio_load` as the prediction target. A `HistGradientBoostingRegressor` was trained as the response model, achieving a test MAE of 12.7878, RMSE of 16.8180, and R^2 of 0.9371. Although this indicates strong offline predictive performance, it did not guarantee stable closed-loop control because the controller later visited states that differed from the random-action training distribution. This motivated the addition of feedback correction and group-aware redistribution in the final controller.

B. RBC Performance

Table IV summarize the performance of the evaluated rule-based controllers under decentralized and centralized control architectures, respectively.

Among all evaluated RBCs, the `PITemperatureController` achieved the best performance in both centralized and decentralized settings. It obtained the lowest CV-RMSE, NMBE, and temperature comfort violation among all RBCs, making it the strongest rule-based baseline. Consequently, it is used as the primary reference controller in subsequent comparisons with the more advanced control approaches.

TABLE IV
PERFORMANCE RESULTS FOR RULE-BASED CONTROLLERS

Decentralized			
Controller	CV-RMSE [%]	NMBE [%]	Comfort Violation [%]
BasicRBC	261.079	166.155	91.39
TemperatureBasedRBC	245.298	-63.133	53.89
OptimizedRBC	220.800	132.824	82.11
BasicBatteryRBC	207.634	89.575	62.14
HourRBC	207.098	115.476	81.90
PITemperatureController	140.523	-54.146	43.64
Centralized			
Controller	CV-RMSE [%]	NMBE [%]	Comfort Violation [%]
TemperatureBasedRBC	266.110	-82.766	82.80
BasicRBC	261.079	166.155	91.39
OptimizedRBC	220.800	132.824	82.11
BasicBatteryRBC	207.634	89.575	62.14
HourRBC	206.770	113.729	81.23
PITemperatureController	158.218	-63.078	63.66

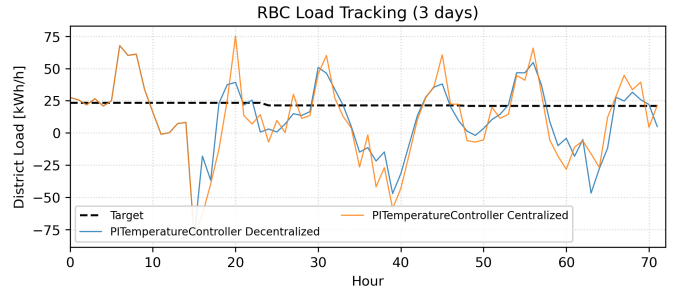


Fig. 6. Best performing centralized and decentralized RBC Load tracking performance over a 3 day period.

Figure 6 shows the aggregated district load generated by the best-performing centralized and decentralized RBCs together with the target load over a three-day period. The figure enables a visual comparison of the load-tracking performance achieved by the two control architectures.

C. MPC Performance

The performance of the three decentralized MPC controllers is summarized in Table V. The evaluated controllers include AL-MPC, HRN-MPC, and CJSS-MPC. Performance was assessed using the CV-RMSE, NMBE, and temperature comfort violation metrics introduced in Section III-I.

TABLE V
PERFORMANCE RESULTS FOR DECENTRALIZED MPCs

Controller	CV-RMSE [%]	NMBE [%]	Comfort Violation [%]
AL-MPC	79.643	10.979	31.42
CJSS-MPC	74.717	-8.083	33.81
HRN-MPC	65.319	18.478	15.33

The results demonstrate noticeable differences in load-tracking accuracy and thermal comfort performance across the three MPC architectures. Figure 7 provides a visual comparison of the aggregated district load profiles produced by each controller over a representative three-day period. The figure illustrates the temporal evolution of the district load relative to the target profile and highlights the different tracking behaviors exhibited by the MPC controllers.

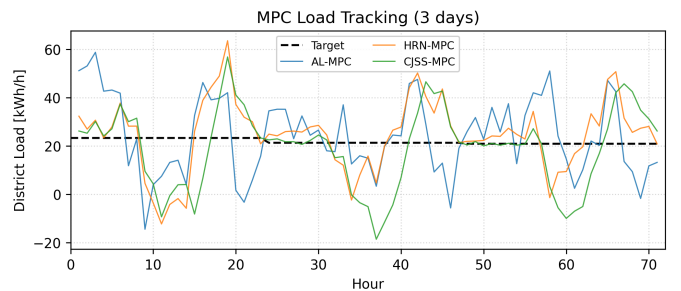


Fig. 7. Load tracking performance of the decentralized MPC controllers over a 3 day period.

D. Initial ML Predictive Controller Performance

The initial ML predictive controller was evaluated in the live CityLearn environment to test whether the trained response model could be used directly for closed-loop district load tracking. The controller achieved an NMBE of 67.8426% and a CV-RMSE of 146.3904%. The large positive NMBE indicates strong systematic tracking bias, while the high CV-RMSE shows that the controller was unable to track the target reliably during live operation. The diagnostic load-tracking plot shows that the predicted next load remained closer to the target than the realised portfolio load, indicating closed-loop distribution shift. These results motivated the final group-aware controller, which adds PI feedback correction and cluster-aware action redistribution.

E. Building Clustering Results

The silhouette-score analysis showed that two clusters achieved the highest clustering score [18]. However, five clusters were selected for the final controller to provide finer control granularity across the 25-building portfolio. This allowed the controller to redistribute actions across more specific building groups while keeping the number of controller groups manageable.

F. Final Group-Aware Controller Performance

The final group-aware controller was evaluated to determine whether PI feedback correction and cluster-aware redistribution improved closed-loop tracking compared with the initial ML predictive controller. The short-term tracking behaviour still showed noticeable hour-to-hour oscillations, indicating that the multi-building system remains difficult to control precisely. However, the feedback and redistribution layers reduced the systematic bias observed in the pure ML controller. This supports the main motivation for the final architecture: prediction alone is not sufficient, but combining prediction with feedback correction and group-aware redistribution produces a more robust controller.

G. Reinforcement Learning Controller Performance

1) *Decentralized SAC Performance*: The NMBE of the decentralized SAC controller evaluated deterministically over one 720-step episode is 0.65 %, the CV-RMSE is 134.4% and the Temperature Comfort Violation is 27.8%. Figure 8 shows the district load profile against the prescribed target. Figure 9 shows the full reward curve over all 100 episodes.

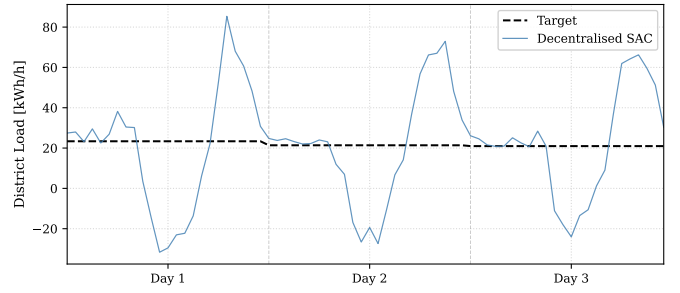


Fig. 8. District aggregate load vs. target profile (3-days), Decentralized SAC. The controller closely tracks the mean target level but exhibits large hour-to-hour deviations.

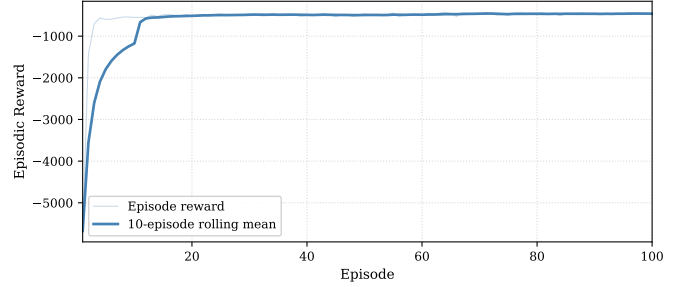


Fig. 9. Episodic reward of the Hier-2 upper-level SAC agent over 100 training episodes.

2) *Hier-2 KPI Performance*: The NMBE of the Hier-2 controller evaluated deterministically over one 720-step episode is 5.1 %, the CV-RMSE is 34.5% and the Temperature Comfort Violation is 27.8%. Figure 10 shows the district load profile against the target.

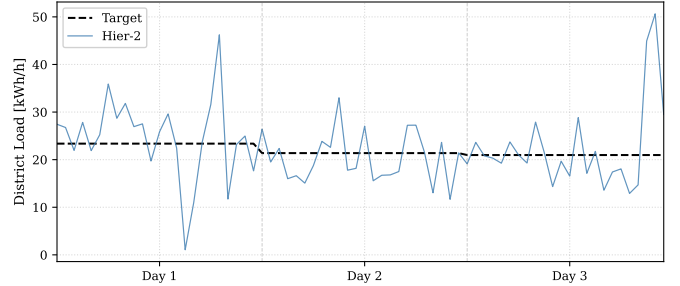


Fig. 10. District aggregate load vs. target profile (3-days), Hier-2.

3) *Hier-2 Variants*: Table VI reports the KPIs of the five Hier-2 variants alongside the original baseline.

TABLE VI
KPI COMPARISON: HIER-2 VARIANTS VS. BASELINE

Controller	CV-RMSE [%]	NMBE [%]	Comfort Violation [%]
Hier-2 (baseline)	34.5	5.08	27.8
Hier-2.3-SOC	34.5	5.25	27.8
Hier-2.1-Solar	35.0	4.77	27.8
Hier-2.5-Asymmetric	38.0	5.85	27.8
Hier-2.2-Extended	40.7	5.07	27.8
Hier-2.4-LargeNet	44.5	4.69	27.8

V. DISCUSSION

The results demonstrate a clear relationship between controller sophistication and district-level load-tracking performance. In general, performance improved as controllers incorporated greater predictive capability, adaptive decision-making, and coordination between buildings. This progression can be observed from rule-based controllers (RBCs), to Model Predictive Controllers (MPCs), and ultimately to machine learning (ML) and reinforcement learning (RL) approaches. The results therefore provide insight into how different control architectures influence load-tracking accuracy, thermal comfort preservation, and overall district-level performance.

The RBC results highlight the limitations of reactive control strategies for district-level load tracking. Controllers such as BasicRBC, HourRBC, and BasicBatteryRBC rely on predefined schedules that do not adapt to future target values, weather conditions, or changing building demand, resulting in large tracking errors. Incorporating feedback improves performance, as demonstrated by the TemperatureBasedRBC and PITemperatureController. By responding to the current thermal state of the building rather than following fixed schedules alone, these controllers achieve more stable HVAC operation and improved energy management decisions. Despite these improvements, all RBCs remain fundamentally limited by their reactive nature. Since future target values and system dynamics are not considered when selecting actions, the controllers cannot proactively prepare battery and cooling resources for upcoming changes in demand. This limitation is particularly important in the Texas scenario, where maintaining a desired district load over an entire day requires anticipation rather than reaction. As a result, RBCs exhibit substantially higher tracking errors than the other control approaches.

The MPC results demonstrate the advantages of predictive control for district-level energy management. By optimizing control actions over a prediction horizon, MPC can anticipate future target changes and proactively allocate battery and cooling resources. Compared to the RBCs, this predictive capability resulted in substantially lower tracking errors and improved comfort performance. Among the evaluated MPC architectures, HRN-MPC achieved the strongest performance. A likely reason for this improvement is its hybrid system identification framework, which combines linear modelling with residual neural network learning. Building thermal behaviour is inherently nonlinear and influenced by multiple interacting factors, including weather conditions, occupancy patterns, and HVAC operation. The hybrid model is therefore better able to capture these dynamics than simpler approximations, resulting in more accurate predictions and improved control decisions. In addition to achieving the lowest tracking error among the MPCs, HRN-MPC also achieved the lowest temperature comfort violation of any controller evaluated in this study. This suggests that the improved prediction accuracy extends beyond load estimation and also enables more precise modelling of building thermal dynamics, allowing cooling actions to be scheduled in a manner that better maintains

indoor temperatures within the desired comfort range. In contrast, AL-MPC relies on local linearizations of a non-linear model, which may become less accurate as operating conditions change. Although CJSS-MPC introduces a richer state representation and district-level coordination, the results suggest that the benefits of additional coordination can be offset by the increased complexity of modelling and predicting a larger set of interacting states. Overall, the MPC results indicate that prediction accuracy is a critical factor in MPC performance, and that increasing model complexity alone does not necessarily lead to improved control performance.

The initial ML predictive controller provides an important intermediate step between the MPC-based approaches and the final group-aware architecture. Although the offline response model performed well on the action-response dataset, the live results showed that offline prediction accuracy did not directly translate into stable closed-loop control. The controller achieved an NMBE of 67.84% and a CV-RMSE of 146.39%, indicating strong systematic bias and poor target tracking.

This behaviour is mainly explained by closed-loop distribution shift. The response model was trained using random-action data, whereas during live operation the controller repeatedly selected actions based on its own predictions. As a result, the states visited during control differed from the training distribution. The diagnostic plots support this: the predicted next load often remained closer to the target than the actual realised load, showing that the model did not fully capture the live environment response.

The final group-aware controller addressed this limitation by combining ML prediction, PI feedback correction, and cluster-aware redistribution. The ML model still proposes an initial action, but the PI layer corrects it using the live tracking error. The corrected action is then redistributed across behaviour-based building clusters, allowing the controller to account for differences in battery state of charge, thermal condition, cooling demand, net load, and solar generation. Mean-action preservation ensures that these cluster-level adjustments do not significantly change the total portfolio action.

Overall, the final group-aware controller improved the main tracking metrics compared with the initial ML predictive controller. NMBE improved from +67.84% to -11.54%, while CV-RMSE decreased from 146.39% to 96.17%. This shows that learned predictive models are useful for district-level control, but are more effective when combined with feedback and coordination mechanisms. However, the remaining CV-RMSE and short-term oscillations indicate that further work is still needed to improve closed-loop tracking reliability.

Finally, the RL results reveal the impact of learned coordination mechanisms on district-level performance. The decentralised SAC controller achieved an NMBE of 0.65%, indicating that the per-building agents collectively tracked the mean district target with high accuracy. However, the CV-RMSE of 134.4% reveals that this accuracy is limited to the temporal mean and does not extend to individual hourly time steps. Without a district-level coordination mechanism, each agent optimises its own building-level reward indepen-

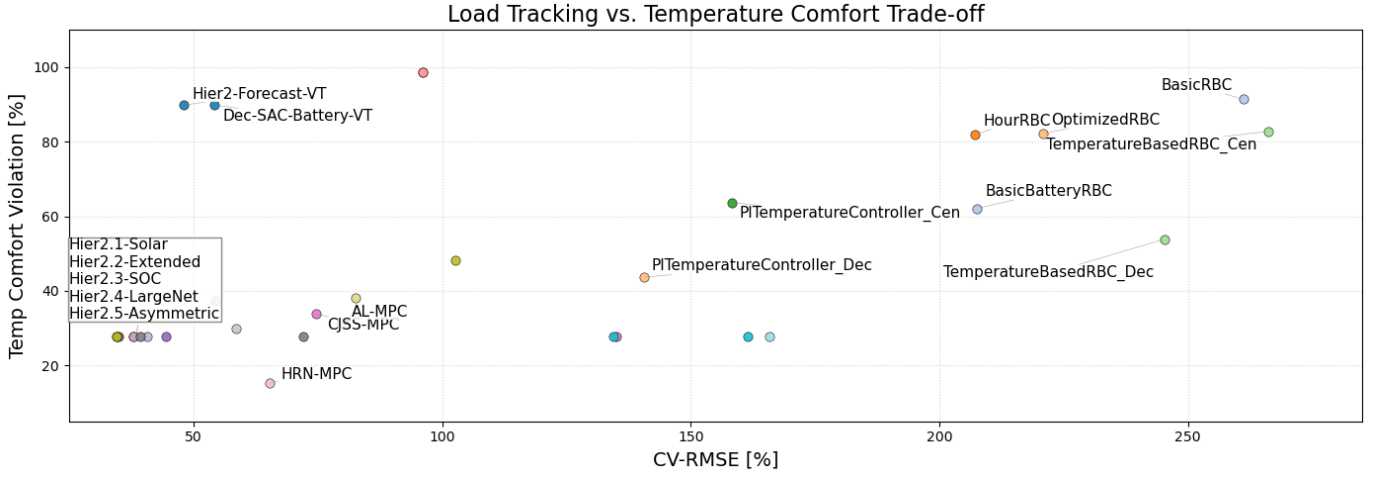


Fig. 11. Trade-off between load-tracking accuracy (CV-RMSE) and temperature comfort violation for all evaluated controllers. Distinct performance clusters can be observed for RBC, MPC, ML-based, and RL-based control approaches.

dently, producing erratic hour-to-hour aggregate behaviour. The temperature comfort violation rate of 27.8% is identical across all RL controllers and is not attributable to battery scheduling. Since cooling-device actions are deactivated in all buildings, indoor temperatures are governed solely by the building thermal model, making this rate a structural property of the dataset rather than a shortcoming of the controller.

The Hier-2 results demonstrate that adding a hierarchical district-level coordinator substantially improves hourly tracking. The CV-RMSE decreased from 134.4% to 34.5%, a reduction of approximately 74%, outperforming the strongest model predictive controller evaluated, HRN-MPC, which achieved 65.3%, without requiring an explicit building model or system identification stage. The improvement can be attributed to the coordinator’s ability to incorporate 6-hour and 12-hour ahead load targets. The marginal increase in NMBE from 0.65% to 5.1% is consistent with the conservative bias introduced by the EMA action smoothing.

None of the five Hier-2 variants produced a meaningful improvement in CV-RMSE within the 100-episode training budget. Extending the forecast horizon (Hier-2.2-Extended) and increasing network capacity (Hier-2.4-LargeNet) produced the largest degradations, suggesting that distant targets add noise to the learning signal and that 100 episodes are insufficient for larger networks to converge. The remaining variants produced negligible or marginal changes, as solar generation is already implicit in the net load signal, SoC distributional statistics offer little additional information given the shared policy structure, and asymmetric action scaling conflicts with the symmetric demand profile. These results indicate that the primary driver of tracking accuracy is the hierarchical architecture combined with multi-step forecasts, rather than observation design, network size, or action scaling.

Taken together, the results provide direct answers to the research questions posed in this study. Figure 11 provides a comprehensive overview of the trade-off between

load-tracking performance and temperature comfort across all evaluated controllers and reveals a clear grouping of controller families. The RBCs occupy the upper-right region of the figure, indicating the weakest overall trade-off, while MPC and ML-based approaches form an intermediate group. The RL-based controllers achieve the strongest load-tracking performance overall. Regarding the first research question, hierarchical coordination proved to be the most effective mechanism for meeting a portfolio-level target profile. The Hier-2 controller achieved a CV-RMSE of 34.5%, compared to 134.4% for the decentralised SAC baseline, 140.5% for the strongest RBC, and 65.3% for the strongest MPC, highlighting the importance of district-level coordination. Regarding the second research question, the results reveal a clear trade-off between load-tracking accuracy and thermal comfort. For example, HRN-MPC achieved the lowest comfort violation among all evaluated controllers, while the action penalty and EMA smoothing used in Hier-2 reduced hourly oscillations at the cost of a small increase in systematic bias. Regarding the third research question, the strongest overall performance was achieved by controllers that combined prediction, adaptation, and coordination. In contrast, controllers that relied solely on reactive heuristics, lacked coordination mechanisms, or operated in open loop consistently achieved weaker overall performance.

VI. CONCLUSION

This study evaluated rule-based, model predictive, machine learning, and reinforcement learning controllers for district-level load tracking in the CityLearn Texas climate scenario. The results showed that controller performance improved as predictive capability and coordination mechanisms were introduced. While RBCs exhibited the weakest tracking performance and MPCs achieved substantial improvements through predictive optimization, the hierarchical reinforcement learning controller (Hier-2) achieved the best overall load-

tracking performance with a CV-RMSE of 34.5%. These findings demonstrate that effective coordination between buildings is a key factor in achieving accurate district-level load shaping and highlight the potential of hierarchical reinforcement learning for smart energy management applications.

The ML-based controller experiments further showed that offline prediction accuracy alone is insufficient for reliable closed-loop control. When the learned response model was used directly, the initial ML predictive controller produced substantial tracking bias, with an NMBE of 67.84% and a CV-RMSE of 146.39%. By adding PI feedback correction and cluster-aware action redistribution, the final group-aware controller reduced the NMBE to -11.54% and the CV-RMSE to 96.17%. This demonstrates that learned predictive models are more effective when combined with feedback and coordination mechanisms that account for building heterogeneity, although further work is still required to reduce short-term oscillations and improve closed-loop tracking reliability.

VII. CODE AVAILABILITY

The source code developed for this project, including the implementations of the controllers, is publicly available on GitHub: https://github.com/Harishk008/annex96_common_exercise_1.git. The repository also contains the supporting scripts used for data processing, feature engineering, controller evaluation, and result generation.

To support experiment tracking and collaboration, Weights & Biases (W&B) website was used throughout the project. Controller configurations, training runs, and evaluation metrics were logged to a shared dashboard, allowing team members to monitor experiments and compare performance using common KPIs and visualizations. The project dashboard is available at: <https://wandb.ai/CityLearn-TeamB/CityLearn?nw=sqjipni0mel>.

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APPENDIX A

INDIVIDUAL CONTRIBUTIONS

A. *Luc van Dijk*

Luc van Dijk contributed primarily to the development and evaluation of the reinforcement learning (RL) controllers presented in this study. He designed, implemented, trained, and analyzed the decentralized and hierarchical RL architectures, including the SAC-based controllers and the Hier-2 coordination framework. His work included controller development, hyperparameter tuning, performance evaluation, and the analysis of the effects of coordination mechanisms, forecast information, and architectural design choices on district-level load-tracking performance.

In addition to the RL implementation, he contributed to the formulation and refinement of the project’s research questions, the development of the literature review, and the formalization of the project methodology. He also participated in the interpretation and presentation of the experimental results, particularly those related to reinforcement learning and district-level coordination strategies.

B. *Harish Kannan*

Harish Kannan’s contribution focused on the development and evaluation of the machine-learning-based control pipeline. He first worked on feature engineering by transforming raw CityLearn observations into structured building-level, portfolio-level, and time-dependent features. These features were used to analyse building behaviour, train the action-response model, and support later clustering and controller development.

Harish then developed and evaluated the initial ML predictive controller and the final group-aware controller. The initial ML controller tested whether the trained response model could directly perform closed-loop target tracking, but its high tracking bias showed the need for feedback correction. To address this, he contributed to the final hierarchical ML–PI–cluster controller, where the ML proposal is corrected using PI feedback and redistributed across behaviour-based building clusters. This improved tracking performance and reduced systematic bias compared with the pure ML controller.

C. *Yusuf Berkan Gokce*

Yusuf Berkan Gokce contributed primarily to the development and evaluation of the Model Predictive Control (MPC) controllers presented in this study. He designed, implemented, and analyzed multiple MPC architectures, including the Adaptive LSTM-Linearized MPC (AL-MPC), Hybrid Residual-NARX MPC (HRN-MPC), and Coordinated Joint State-Space MPC (CJSS-MPC). His work included the development of the system identification pipelines, formulation of the optimization problems, controller implementation, performance evaluation, and comparison of the proposed controllers.

In addition, he evaluated the rule-based controllers (RBCs) already available within the CityLearn environment and analyzed their performance for the Texas district scenario. These controllers served as baseline references against which the more advanced MPC, machine learning, and reinforcement learning approaches were compared. He also developed a shared Weights & Biases (W&B) dashboard that enabled team members to log results, monitor experiments, and compare controller performance using common KPIs and visualizations.